airbag housing and tabs on the retainers are engaged in slots within the upper and lower trim cover locking blocks, securely locking the cover into place. The stamped metal retainers are secured to the four airbag inflator mounting studs with nuts.

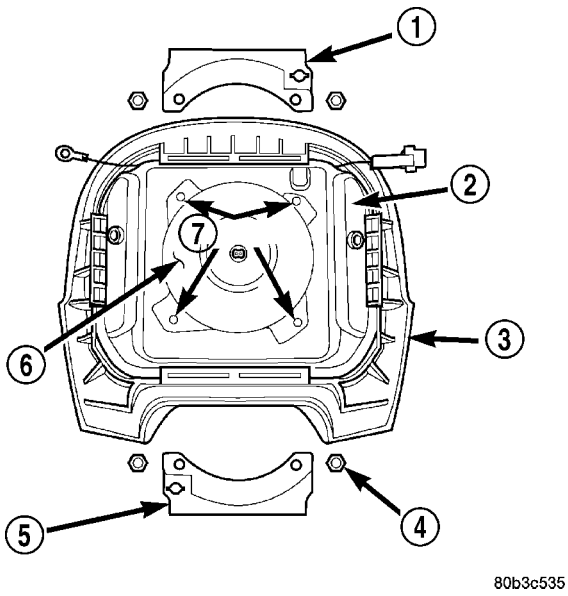

Fig. 21 Driver Airbag Trim Cover

1 - UPPER RETAINER
2 - AIRBAG HOUSING
3 - TRIM COVER
4 - NUT (4)
5 - LOWER RETAINER
6 - INFLATOR
7 - STUD (4)

The resistive membrane-type horn switch is secured within a plastic tray that is inserted into a pocket or pouch sewn onto the airbag cushion retainer strap, between the trim cover and the folded airbag cushion. The horn switch ground pigtail wire has an eyelet terminal connector that is captured on the upper right inflator mounting stud between the inflator and the upper trim cover retainer. The horn switch feed pigtail wire has a white, molded plastic insulator that is secured by an integral retainer to a mounting hole located in the upper trim cover retainer near the upper left corner on the back of the airbag housing, and is connected to the vehicle electrical system through a dedicated pigtail wire and connector from the clockspring.

The airbag used in this model is a Next Generation-type that complies with revised federal airbag standards to deploy with less force than those used in some prior models. A radial deploying fabric cushion with internal tethers is used. The airbag inflator is a conventional pyrotechnic-type unit that is secured by four hex nuts to four studs that extend through the back of the stamped metal airbag housing from the airbag cushion retainer ring. A keyed connector receptacle on the driver airbag inflator connects the inflator initiator to the vehicle electrical system through a yellow-jacketed, two-wire pigtail harness of the clockspring. The driver airbag cannot be repaired, and must be replaced if deployed or in any way damaged. The driver airbag trim cover and the horn switch are available individually, and may be disassembled from the driver airbag for service replacement.

OPERATION

The driver airbag is deployed by electrical signals generated by the Airbag Control Module (ACM) through the driver airbag squib circuit to the initiator in the airbag inflator. When the ACM sends the proper electrical signal to the initiator the electrical energy generates enough heat to initiate a small pyrotechnic charge which, in turn ignites chemical pellets within the inflator. Once ignited, these chemical pellets burn rapidly and produce a large quantity of inert gas. The inflator is sealed to the back of the airbag housing and a diffuser in the inflator directs all of the inert gas into the airbag cushion, causing the cushion to inflate. As the cushion inflates, the driver airbag trim cover will split at predetermined breakout lines, then fold back out of the way along with the horn switch unit. Following an airbag deployment, the airbag cushion quickly deflates by venting the inert gas towards the instrument panel through the porous fabric material used to construct the back (steering wheel side) panel of the airbag cushion.

Some of the chemicals used to create the inert gas may be considered hazardous while in their solid state before they are burned, but they are securely sealed within the airbag inflator. Typically, all potentially hazardous chemicals are burned during an airbag deployment event. The inert gas that is produced when the chemicals are burned is harmless. However, a small amount of residue from the burned chemicals may cause some temporary discomfort if it contacts the skin, eyes, or breathing passages. If skin or eye irritation is noted, rinse the affected area with plenty of cool, clean water. If breathing passages are irritated, move to another area where there is plenty of clean, fresh air to breath. If the irritation is not alleviated by these actions, contact a physician.

REMOVAL

The following procedure is for replacement of a faulty or damaged driver airbag. If the airbag is faulty or damaged, but not deployed, review the recommended procedures for handling non-deployed supplemental restraints. (Refer to 8 - ELECTRICAL/ RESTRAINTS - STANDARD PROCEDURE - HANDLING NON-DEPLOYED SUPPLEMENTAL RESTRAINTS). If the driver airbag has been deployed, review the recommended procedures for service after a supplemental restraint deployment